



Department of Creative Technologies

Software Engineering



Course Title	Natural Language Processing		
Course Code	SE 454	Offered In	Fall 2023
Pre-Requisites	--	Credit Hours	3 – 0 – 3
Cluster Head	Dr. Imran Ihsan	Course Type	SE Elective
Faculty / Dept.	FCAI – Department of Creative Technologies (Software Engineering)		
Instructor Name	Dr. Imran Ihsan	GCR Code	
Designation	Associate Professor	Office	3 RD Floor, C – Block
Email	imranihsan@mail.au.edu.pk	Office Hours	
Offered to	BS – Software Engineering	Semester	VII
Class Days	Tuesday	Friday	
Class Time	1440 – 1600	0800 – 0920	
Class Location	C – 404	C – 401	

COURSE INFORMATION / DESCRIPTION OF THE COURSE

This course introduces the theory and methodology of natural language understanding and generation. Topics include stemming, lemmatization, parts of speech tagging, parsing, and machine translation. Employing specialized libraries, students develop applications for topic modeling, sentiment analysis, and text summarization.

LEARNING RESOURCES

Speech and Language Processing By: Dan Jurafsky and James H. Martin, 2023.

Reference Books

Natural Language Understanding in a Semantic Web Context, By Caroline Barrière, Springer, 2017

COURSE LEARNING OUTCOMES

At the end of this course, students will be able to:

No	Description	Bloom's Taxonomy	
1	Extract information from text automatically using concepts and methods from NLP including stemming, n-grams, POS tagging, and parsing.	C2	Understand
2	Analyze the syntax, semantics, and pragmatics of a statement written in a natural language.	C2	Understand
3	Apply machine learning algorithms to natural language processing.	C3	Apply
4	Write scripts and applications in Python to carry out natural language processing using libraries such as NLTK, Gensim, and spaCY.	C6	Create

C1 – Remember C2 – Understand C3 – Apply C4 – Analyze C5 – Evaluate C6 – Create

PROGRAM OUTCOMES

PLO	Program Learning Objectives	CLO	Assessed By
1	Academic Education	1, 2	Assignments
2	Knowledge for Solving Computing Problems	3	Assignments
3	Problem Analysis	3	Assignments, Exam
4	Design / Development of Solutions	3, 4	Assignments, Project
5	Modern Tool Usage	4	Project
6	Individual and Teamwork	--	--
7	Communication	--	--
8	Computing Professionalism and Society	--	--
9	Ethics	--	--
10	Lifelong Learning	--	--

ASSESSMENT COMPONENTS

Assessment Method	Description	Weight	Aligned CLOs
Assignments	08 Practical Assignments	16%	1, 2, 3
Group Project	End Semester Project	10%	4
Class Participation	Based on behaviors that students engage in class	04%	--
Midterm Exam	In Week 08, As per AU Policy	25%	1, 2, 3
Final Exam	In Week 15/16, As per AU Policy	45%	2, 3, 4

TENTATIVE COURSE OUTLINE			
Week	Date	Topics & Reading	Related CLOs
1	16 Sep – 20 Sep	Language And Morphology - Linguistic Basics - Morphology - Evaluation, Precision and Recall - Regular Expressions	
2	23 Sep – 27 Sep	Sentiment Analysis with Logistic Regression - Sentiment Analysis & Feature Extraction - Preprocessing - Logistic Regression - Preprocessing using Python - Visualizing Word Frequencies - Logistic Regression Model	CLO – 2
3	30 Sep – 04 Oct	Sentiment Analysis with Naïve Bayes - Probability and Bayes' Rule - Naïve Bayes for Sentiment Analysis - Laplacian Smoothing - Log Likelihood - Naïve Bayes – Training and Testing - Applications of Naïve Bayes - Naïve Bayes – Problems - Visualizing Naïve Bayes	CLO – 2
4	07 Oct – 11 Oct	Vector Space Models - Vector Space Models – Introduction - Euclidean Distance and Cosine Similarity - Principal Component Analysis – PCA Algorithm - Linear Algebra in Python with NumPy - Manipulating Word Embeddings - Implementing PCA	CLO – 3
5	14 Oct – 18 Oct	K-Nearest Neighbors – KNN - Machine Translation - Nearest Neighbors and Hashing - Locality Sensitive Hashing - Document Search - Vector Manipulation in Python - Hash Functions and Multiplanes	CLO – 2
6	21 Oct – 25 Oct	Dynamic Programming - Machine Translation - Nearest Neighbors and Hashing	CLO – 2

		• Locality Sensitive Hashing • Document Search • Vector Manipulation in Python • Hash Functions and Multiplanes	
7	28 Oct – 01 Nov	Part of Speech Tagging and Hidden Markov Model • Part of Speech • Part of Speech (POS) Tagging • Transition Probabilities • Hidden Markov Model • Emission Probabilities • Calculate Transition Probabilities • Calculate Emission Probabilities • Viterbi Algorithm	CLO – 3
8	04 Nov – 08 Nov	Midterm	
9	11 Nov – 15 Nov	Generative Language Models • N-Gram and Probabilities • N-Gram Language Model • Perplexity • Out of Vocabulary Words • Smoothing & Interpolation	CLO – 2
10	18 Nov – 22 Nov	Word Embeddings with Neural Networks • Word Representations • Word Embeddings • Continuous Bag-of-Words Model • Sliding Window of Words in Python • Transforming Words into Vectors	CLO – 2
11	25 Nov – 29 Dec	Word Embeddings with Neural Networks • Architecture of CBOW Model • CBOW Model – Activation Function • CBOW Model – Cost Function • Backpropagation and Gradient Descent • Extracting Word Embedding Vectors • Intrinsic / Extrinsic Evaluation	CLO – 1
12	02 Dec – 06 Dec	Neural Networks for Sentiment Analysis • Neural Networks and Forward Propagation • Structure of Sentiment Analysis • Neural Networks using Trax	CLO – 1
13	09 Dec – 13 Dec	Neural Networks for Sentiment Analysis • Classes, Subclasses and Inheritance • Dense and ReLU Layers	CLO – 1

		Serial and Other Layers	
14	16 Dec – 20 Dec	RNN For Language Modelling - Traditional Language Models - Recurrent Neural Networks and Its Applications - Cost Function for RNNs - Recurrent Units - Deep and Bi-Directional RNNs	CLO – 1
15	23 Dec – 27 Dec	LSTM and Named Entity Recognition - RNNs and Vanishing Gradients - LSTM Architecture - Named Entity Recognition - Data Processing and Computing Accuracy	CLO – 4
16	30 Dec – 03 Jan	Final Exam	
17	06 Jan – 10 Jan	Final Exam	

PRACTICAL / PROGRAMMING WORK / TOOLS			
Python, NLTK, Gensim, SpaCy, Trax, Python Libraries for Machine Learning			
CLASS TIME SPENT ON			
Theory	Problem Analysis	Solution Design	Social & Ethical Issues
35%	35%	25%	05%
ORAL AND WRITTEN COMMUNICATIONS			
Written Reports	Average Report Size	Oral Presentations	Presentation Duration
1	5 Pages	1	10 Minutes

COURSE PLAGIARISM POLICY
Plagiarism in project or mid/ final exam will result in F grade in the course.
Plagiarism in an assignment will result in zero marks in the whole assignments category.

COURSE OUTLINE APPROVAL		
Course Instructor	Cluster Head	Program Manager